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Power requirements for olive mechanical harvesting using trunk shaker

Mohamed Ghonimy[1]✉, Mohamed M. Ibrahim[2]✉, Haytham S. Helmy[2] & Ahmed Alzoheiry[1,3]

A mathematical analysis was conducted to predict the power requirements of the olive trunk shaker. The predicted required power of the olive trunk shaker is influenced by shaking displacement, fruit detachment force, fruit mass, stem length, damping ratio, unbalanced mass of the tree shaker, and exciting shaking frequency. The Olive trunk shaker was tested in the field at five attachment heights, 400, 500, 600, 700, and 800 mm, and six levels of olive trunk diameter: 78, 84, 101, 143, 158, and 186 mm. The results indicated that the theoretical shaking power was 12.8 kW while the actual required shaking power ranged from 7.1 to 17.8 kW, with an average of 12.6 kW due to variations in tree trunk diameters and different attachment heights. Multiple regression using the least squares method was employed to establish a statistical relation between the theoretical equation and the actual required power as a function of the clamp attachment height and trunk diameter. The regression coefficient of the predicted equation was 93.6% for attachment height values between 400 and 800 mm and tree trunk diameter values between 78 and 186 mm, with a mean absolute error of the prediction model was 0.5009 kW.

Keywords Fruit, Modeling, Olive, Power, Trunk shaker

Mechanical harvesting of olives is one of the most essential agricultural operations that affect the cost of the final product. The mechanical harvesting of olives involves two primary methods; shaking or combing the tree, as described by Nasini and Proietti[1]. Farmers tend to favor shaking techniques over combing due to their lesser impact on tree health. Bernardi et al.[2] identified several factors influencing the efficiency of mechanical shaking harvests for tree fruits, including frequency, eccentricity, shaking direction, fruit size, and the ratio of detachment force to fruit weight. Resonance energy transfer can disperse mechanical energy throughout the entire tree during olive harvesting by shaking the frequency bands overlapping between primary branches, within branches, and throughout the tree. This method dissipates mechanical energy more effectively than in a tree with stiff branches because it is less concentrated on the trunk and roots. Vibration frequency significantly affects energy distribution across tree components like the trunk, branches, and canopy. High frequencies concentrate energy near the shaking point, reducing transmission to the branches and canopy, which may lower fruit detachment efficiency. In contrast, medium and lower frequencies distribute energy more evenly, improving detachment but potentially increasing vibrations in roots and soil, affecting tree health[3]. The response varies with tree characteristics such as wood stiffness and branch length, making it crucial to adjust the frequency according to the specific olive variety to optimize efficiency and minimize damage[4]. Harvesting table olives is increasingly done with trunk shakers[5]. A straight trunk that is between 0.6 and 1 m below the canopy is necessary for trunk-shaking harvesters[6]. Trunk shakers have been the subject of numerous studies in the context of olive and apple harvesting. Several impact energy levels were used to determine the tree responses to impact trunk shaking[7,8]. According to recent research, shaken fruit trees may experience very large damping losses, depending on how high the shaker is linked to the tree trunk[9,10]. Studies that were conducted on the branch vibrators showed that the actual energy consumed converged by up to 95% of the theoretical calculations for energy estimation due to the majority of the net power was dissipated by the damping losses[11–13]. According to experimental observations, trunk shaker power consumption strongly depends on attachment height. Simultaneously, many shaker arrangements prohibit the utilization of elevated attachment points even though this would be a relatively

[1]Department of Agricultural and Biosystems Engineering, College of Agriculture and Food, Qassim University, P.O. Box 6622, Buraydah, Al-Qassim 51452, Saudi Arabia. [2]Department of Agricultural Engineering, Faculty of Agriculture, Cairo University, Giza, Egypt. [3]Department of Natural Resources and Agricultural Engineering, Faculty of Agriculture, Damanhour University, Damanhour, Egypt. ✉email: m.elsayed@qu.edu.sa; mohamed.ibrahim@agr.cu.edu.eg

straightforward method of lowering energy usage. Ghonimy[14] came up with a formula to estimate the ideal amplitude at which a tree shaker should shake. The developed equation establishes a relationship between the shaking amplitude and the fruit mass ratio, pulling force, shaking frequency, stem length, and damping ratio. A fruit tree is a vibrating system comprising several vibrating components, including the trunk, branches, limbs, and root-soil mass[15]. When the shaking frequencies of a shaker get close to the natural frequency of the tree, the maximum amount of fruit is removed. However, the ideal shaking frequency differs for every single tree because the natural frequency of a tree depends on its size, morphology, age, wood qualities, leaf density, and moisture content[16,17]. According to Ghonimy et al.[18], the average maximum bending stress influencing the limb's deflection was 196.6 mm, and the maximum bending stress was 16.5 MPa. This study aimed to predict the required power of the trunk shaker to harvest olive trees.

Materials and methods

This study used an analytical methodology to investigate the impact of various parameters on the trunk shaker power of olive trees. The initial step was to perform a mathematical analysis in order to determine the mathematical relationship between the many elements influencing the trunk shaker power. Secondly the experimental work aimed to verify the accuracy of the theoretical part of the study.

The maximum required power for an olive trunk shaker (Eq. 1) depends on amplitude or trunk shaking displacement (x), frequency (ω), and maximum force (F_{max}) acting on the tree trunk.

$$P_r = x.\omega \ .F_{max} \tag{1}$$

Shaking displacement

The illustrated tree model utilized in this study is depicted in Fig. 1. The overall displacement (x) at the attachment point, as described in Eq. (2), comprises three components: x_1 represents the lateral motion of the tree, m; x_2 indicates the tilting motion, m; and x_3 accounts for the elastic deformation of the tree trunk, m.

$$x = x_1 + x_2 + x_3 \tag{2}$$

The maximum force acting on the tree trunk

The movement of the olive trunk is rather complex. Its biological makeup is characterized by non-uniformity and an extensive range of degrees of freedom. As a result, the trunk was viewed as a stiffness member with internal damping, and the trunk system was examined as one degree of freedom. To predict the maximum force acting on the tree trunk, the olive trunk needs the following characteristics: the total spring constant (k), N·m^{-1}, trunk damping ratio (ζ); and natural frequency of olive limbs (ω_n), rad·s^{-1}.

The total spring constant (k) of the olive trunk was calculated from the Eq. (3).

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} \tag{3}$$

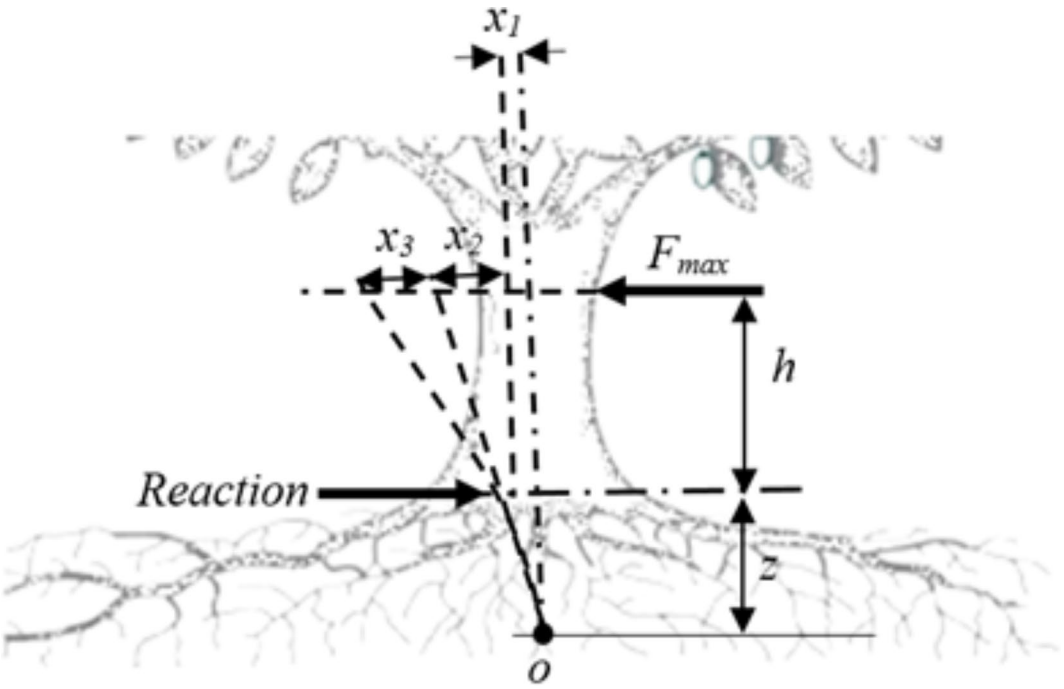


Fig. 1. Movement types of olive tree trunk under shaking.

where k_1 is the lateral motion spring constant, N·m^{-1}; k_2 is the tilting motion spring constant, N·m^{-1}; and k_3 is the tree trunk's elastic spring constant, N·m^{-1}. Applying the theory of beam defection, the trunk k_3's elastic spring constant is computed, but the other two need to be found by experimentation. Because soils behave in such a highly plastic manner, the spring constants k_1 and k_2 can be neglected and will not have any impact. Therefore, the equation is simplified to $\frac{1}{k} \approx \frac{1}{k_3}$, which implies that $k \approx k_3$.

Therefore, the total spring constant (k) of the olive trunk was 3.166 kN·m^{-1}.

The maximum force acting on the tree trunk (F_{max}) was determined from Eq. (4):

$$F_{max} = m_u\, x\, \omega^{\,2} \tag{4}$$

Where F_{max} is the maximum force acting on the tree trunk, N; m_u is the unbalanced mass of the tree shaker, kg; x is the crank mechanism's eccentricity, m; and ω is the shaking frequency in rad·s^{-1}.

General assumptions and simplifications for mathematical analysis

1. The spring constants k_1 and k_2 can be neglected.
2. The fruits are treated as homogeneous with a uniform density.
3. The shaking direction is assumed to be horizontal.

Shaking frequency

When harvesting fruit mechanically by shaking, the tree branch receives external vibration energy to remove the fruit. Various forms of energy are created from this energy. The method for mathematical analysis described by Eq. (5) in La-Grange[19] was utilized as the suitable approach.

$$\frac{d}{dt}\left(\frac{\partial KE}{\partial \theta^{\bullet}}\right) - \frac{\partial KE}{\partial \theta} + \frac{\partial PE}{\partial \theta} + \frac{\partial D}{\partial \theta^{\bullet}} = \frac{\partial W}{\partial \theta} \tag{5}$$

Where KE is the kinetic energy, N·m; $\theta^{\bullet}$ is the angular velocity of the fruit stem, rad·s^{-1}; θ is the fruit stem's angular displacement from its vertical position, rad; PE is the potential energy, N·m; D is the minuscule amount of energy lost per unit of time, N·m·s^{-1}; W is the external work, N·m.

La Grange equation is regarded as the energy balance equation for every vibrating system. The kinetic energy used to move the fruit, the potential energy used to change the fruit's location, and the internal and external dissipated energy are the various converted energy forms that were used to express the external work in this equation.

The effect of various influencing elements on the shaking amplitude that results in fruit separation must be formally expressed by solving the partial differential equation using an appropriate sophisticated approach. The partial differential Eq. (5) is transformed into an ordinary differential equation by applying the complex approach.

a. **The amount of kinetic energy (*KE*) used to move olive fruit**.

From Eq. (6), the kinetic energy (KE) used by a fruit in motion is dependent on its mass (m) and speed (v).

$$KE = \frac{1}{2}mv^2 \tag{6}$$

Referring to Figs. 2,

$$v^2 = (\ell\,\theta^{\bullet} cos\theta\, + x^{\bullet})^2 + (\theta^{\bullet}\ell\, sin\theta\,) \tag{7}$$

Where ℓ is stem length, m; x is the displacement caused by shaking, m; θ is the deviation of an angle from a vertical position, rad.

Using Eqs. (7) and (6) gives:

$$KE = \frac{m}{2}\left[(x^{\bullet})^2 + 2\ell x^{\bullet}\theta^{\bullet} cos\theta\, + \ell^2\theta^{\bullet 2}\right] \tag{8}$$

b. **Potential energy (*PE*)**.

The potential energy (PE) expended in changing the position of the fruit was calculated using Eq. (9). The fruit mass (m) and the vertical distance (z) between the starting and finishing locations are used to compute it.

$$PE = mgz \tag{9}$$

Referring to Fig. 2, $z = \ell - \ell\cos\theta$, thus

$$PE = mg\ell\,(1 - cos\theta\,) \tag{10}$$

c. **The minuscule amount of energy lost per unit of time (*D*)**.

D was found from Eq. (11)[20].

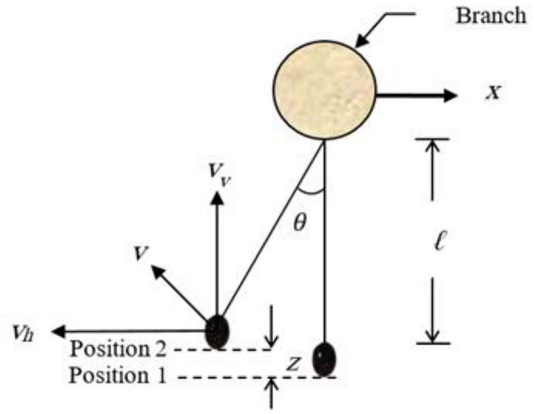


Fig. 2. The movement of the fruit stem when shaking.

$$D = \frac{1}{2} C \theta^{\bullet 2} \tag{11}$$

Where C is the damping coefficient, N·m·s; $\theta^{\bullet}$ is the angular speed, rad·s^{-1}.

d. **The external work (W).**

The W can be calculated from Eq. (12) as follows,

$$W = mgx \tag{12}$$

Where m is the total mass of limb and fruits, kg; x is the limb displacement, m.

Thus,

$$\frac{d}{dt}\left(\frac{\partial KE}{\partial \theta^{\bullet}}\right) = m\,\ell\left(\frac{d^2x}{dt^2}\cdot\cos\theta - \frac{dx}{dt}\cdot\frac{d\theta}{dt}\cdot\sin\theta + \ell\frac{d^2\theta}{dt^2}\right) \tag{13}$$

$$\left(\frac{\partial KE}{\partial \theta}\right) = \frac{\partial}{\partial\theta}\left[\frac{m}{2}\left\langle\left(\frac{dx}{dt}\right)^2 + 2\ell\cos\theta\frac{dx}{dt}\cdot\frac{d\theta}{dt} + \ell^2\left(\frac{d\theta}{dt}\right)^2\right\rangle\right] = -m\,\ell\frac{dx}{dt}\cdot\frac{d\theta}{dt}\cdot\sin\theta \tag{14}$$

$$\frac{\partial PE}{\partial\theta} = \frac{\partial}{\partial\theta}\left[mg\ell\,(1-\cos\theta)\right] = mg\ell\sin\theta \tag{15}$$

$$\frac{\partial D}{\partial\theta^{\bullet}} = \frac{\partial}{\partial\theta^{\bullet}}\left[\frac{1}{2}C\left(\frac{d\theta}{dt}\right)^2\right] = C\frac{d\theta}{dt} \tag{16}$$

$$\frac{\partial W}{\partial\theta} = \frac{\partial}{\partial\theta}\left(M.g.x\right) = 0 \tag{17}$$

Using Eqs. 5, 13, 14, 15, 16 and 17 gives:

$$\frac{d^2x}{dt^2}\cos\theta + \frac{d^2\theta}{dt^2}\ell + g\sin\theta + \frac{d\theta}{dt}\cdot\frac{C}{m\ell} = 0 \tag{18}$$

For small values of θ in the second-order differential equation[19] (Eq. 18), $cos\theta = 1$ and $sin\theta = \theta$

$$\therefore \frac{1}{\ell}\cdot\frac{d^2x}{dt^2} + \frac{d^2\theta}{dt^2} + \frac{g}{\ell}\cdot\theta + \frac{C}{m\ell^2}\cdot\frac{d\theta}{dt} = 0 \tag{19}$$

$$\because \frac{C}{m\ell^2} = 2\zeta\cdot\sqrt{g/\ell} \tag{20}$$

Where C is the damping coefficient, N·m·s; m is the mass of fruit, kg; ℓ is the length of stem, m; ζ is the damping ratio.

Using Eqs. 18 & 19 and solving by complex method give[19]:

$$-\omega^2\cdot e^{i\phi} + 2i\omega\zeta\cdot\sqrt{g/\ell}\cdot e^{i\phi} + \frac{g}{\ell}\cdot e^{i\phi} = \frac{\omega^2 x}{\ell\cdot\vartheta} \tag{21}$$

$\because e^{i\phi} = \cos\phi + i \cdot \sin\phi$

$$-\omega^2 (\cos\ \phi + i \cdot \sin\ \phi) + 2i\omega\zeta\sqrt{g/\ell} \cdot (\frac{g}{\ell} \cdot (\cos\ \phi + i \cdot \sin\ \phi) + (\cos\ \phi + i \cdot \sin\ \phi)) = \frac{\omega^2 x}{\ell \cdot \vartheta} \tag{22}$$

The real part of Eq. 22:

$$\left(-\omega^2 + \frac{g}{\ell}\right)\cos\phi - 2\omega\sqrt{\frac{g}{\ell}}\zeta\sin\phi = \frac{\omega^2 x}{\ell \cdot \vartheta} \tag{23}$$

The imaginary part of Eq. 22:

$$-\omega^2 i\sin\phi + 2i\omega.\zeta \cdot \sqrt{g/\ell} \cdot \cos\phi + \frac{g}{\ell} i\sin\phi = 0$$

Dividing by $\boldsymbol{i}$ gives:

$2\omega.\zeta \cdot \sqrt{g/\ell} \cdot \cos\phi = \omega^2 \sin\phi - \frac{g}{\ell}\sin\phi$

$$\sin\phi = \frac{2\omega.\sqrt{\frac{g}{\ell}} \cdot \zeta}{\sqrt{\left(2\omega\sqrt{\frac{g}{\ell}} \cdot \zeta\right)^2 + \left(\omega^2 - \frac{g}{\ell}\right)^2}} \tag{24}$$

$$\cos\phi = \frac{\omega^2 - \frac{g}{\ell}}{\sqrt{\left(2\omega.\sqrt{\frac{g}{\ell}} \cdot \zeta\right)^2 + \left(\omega^2 - \frac{g}{\ell}\right)^2}} \tag{25}$$

Using Eqs. (23), (24) and (25) gives:

$$-\sqrt{\left(2\omega.\sqrt{\frac{g}{\ell}} \cdot \zeta\right)^2 + \left(\omega^2 - \frac{g}{\ell}\right)^2} = \frac{\omega^2 x}{\ell\vartheta} \tag{26}$$

Thus, the angular motion amplitude (ϑ) is:

$$\therefore\ \vartheta = -\frac{\omega^2 x}{\ell * \sqrt{\left(2\omega.\sqrt{\frac{g}{\ell}} \cdot \zeta\right)^2 + \left(\omega^2 - \frac{g}{\ell}\right)^2}} \tag{27}$$

Multiplying the Eq. (27) on both sides by ω.

$$\text{Thus, } |\ \omega\ \vartheta\ | = \frac{\omega^3 x}{\ell * \sqrt{\left(2\omega.\sqrt{\frac{g}{\ell}} \cdot \zeta\right)^2 + \left(\omega^2 - \frac{g}{\ell}\right)^2}} \tag{28}$$

Meanwhile, the necessary inertia force to remove each fruit is equal to $mass \times acceleration$. Thus,

$$F = (\omega\ \vartheta\ )^2 \ell \cdot\ m \tag{29}$$

Where F is fruit detachment force, N; $(\omega * \vartheta)$ is the maximum frequency, rad·s^{-1}.

Equation (29), when substituted from Eq. (28), gives:

$$F = \frac{\omega^6 x^2 m}{\ell\ \left[\left(2\omega\sqrt{\frac{g}{\ell}} \cdot \zeta\right)^2 + \left(\omega^2 - \frac{g}{\ell}\right)^2\right]} \tag{30}$$

Thus, arranging the Eq. (30) gives:

$$\omega\ = \left\{\frac{1}{x^2} \cdot \frac{F}{m} \cdot\ \ell\left[\left(2\omega\sqrt{\frac{g}{\ell}}\zeta\right)^2 + \left(\omega^2 - \frac{g}{\ell}\right)^2\right]\right\}^{1/6} \tag{31}$$

To match the calculated value of frequency from Eq. (31) with those measured at the trunk, a correction factor (cf.) was calculated as the ratio between the measured frequency transmitted to the trunk and the theoretically calculated frequency from Eq. (31). The frequency transmitted to the olive tree trunk was measured using an accelerometer (model 4506, Brüel & Kjær, Nærum, Denmark) mounted at 5 cm above the shaker attachment point.

Using Eqs. 1, 4, 30, and cf., the required power can be calculated from Eq. (32):



Fig. 3. The olive trunk shaker.

Specification	Value
Length, mm	1400
Width, mm	900
Height, mm	1100
Max. clamping diameter, mm	300 mm (500 mm with optional large shaker head)
Shaking performance, trees·h^{-1}	150 to 200
Weight, kg	490
Unbalanced mass, kg	135
Eccentricity, mm	22
Telescopic, mm	2×1000 mm (Option: 3×1000 mm)
Pump performance, l·min^{-1}	Approx. 40
Particularities	Control panel
	PTO shaft drive at 540 rpm
	Hydraulics on board

Table 1. Technical specifications of the olive trunk shaking machine.

$$P_r = 0.001 \times \left\{ cf.\frac{1}{x^2} \cdot \frac{F}{m} \cdot \ell \left[\left(2\omega \sqrt{\frac{g}{\ell}\zeta} \right)^2 + \left(\omega^2 - \frac{g}{\ell} \right)^2 \right] \right\}^{1/6} .m_u\, x^2\, \omega^2 \tag{32}$$

Where P_r is the predicted required power, kW; cf. is the correction factor, dimensionless; x is shaking displacement, m; F is the fruit detachment force, N; m is the olive fruit mass, kg; ℓ is the stem length, m; ζ is damping ratio; m_u is the unbalanced mass of trunk tree shaker in kg; and ω is the exciting shaking frequency in rad·s^{-1}.

Experimentation is the second part of the design of this scientific approach. This experimental effort is regarded as a mean for validating Eq. (32).

Field experimental work

The olive trunk shaker (Feucht Obsttechnik, FSI – Plantation shaker, Germany, Fig. 3) was tested in the Egyptian governorate of Beheira in Wadi El Natroun. Table 1 contains a list of the machine's specifications.

Measurements and calculations

Plant characteristics of Picual olives

Picual olive fruits were visually sorted into three categories based on their ripening stages, as defined by Guzmán et al.[21], relying on the color of both the skin and flesh. These categories included: (a) Full-mature, indicated by yellow-green skin color; (b) Half-ripe, where more than half of the skin displayed hues of red, purple, or black; and (c) Full-ripe, characterized by entirely purple or black skin with purple flesh reaching the pit (Fig. 4). With different maturity levels, 50 olive fruit-stem combinations were chosen randomly to measure fruit length, diameter, mass (m), detachment force (F), and calculate the F/m ratio.

Measurements were taken using a vernier caliper for fruit length, diameter, and stem length, with an accuracy of ±0.1 mm. Detachment force (F) was determined using a digital force gauge with an accuracy of ±0.01 N, while fruit mass was measured using a digital balance with an accuracy of ±0.01 g.

Damping ratio (ζ)

The damping ratio (ζ) was calculated from Eq. (33)[14].

$$\zeta = \sqrt{1 - \frac{\omega_d^2 * \ell}{g}} \tag{33}$$

Where ω_d is the damping frequency $= {}^{2\pi}/_{t}$, rad·s^{-1}; t is the duration of one cycle, s.

A test was undertaken to determine the duration of a single cycle (t). The olive-stem system was positioned vertically on the plate, and the fruits were shifted horizontally. Using a timer, the duration of one complete cycle was measured.

Field experiments

The olive trunk shaker (Fig. 3) was operated on 12-year-old olive trees. The diameters of the olive tree trunk were measured at five heights: 400, 500, 600, 700, and 800 mm from the soil surface. The trunk diameter values were 78, 84, 101, 143, 158, and 186 mm. Thus, the trunk shaker was operated at six trunk diameters (78, 84, 101, 143, 158, and 186 mm) and five levels of attachment height (400, 500, 600, 700, and 800 mm). The trunk shaker was operated at a frequency of 18 Hz[3] and 12 s·tree^{-1} of operating time[22]. The selection of the 18 Hz frequency was driven by its potential to enhance the shaking process by promoting a broader energy distribution across the tree. Medium frequencies, such as 18 Hz, are known to facilitate more effective fruit detachment by transferring energy throughout the tree's structure, including branches and canopy. Thus, this frequency can improve the efficiency of the shaking process[3].

For any treatment, the actual shaking power (P_a) was measured, and fruit removal efficiency (η_{fr}) was calculated. To measure the P_a, an oil flow sensor (Flo-tech, FSC-1000, Flotech Solutions Limited, Stockport, UK) and a pressure sensor (Parker, SCP-600, Parker Hannifin Corp., Cleveland, OH, USA) were used to measure the oleo-hydraulic power before the hydraulic motor that moves the eccentric mass. The actual shaking power (P_a) required to operate the olive trunk shaker was calculated from Eq. (34).

$$P_a = \frac{Q \times P}{600} \tag{34}$$

Where *Pa* is the actual shaking power, kW, Q is the hydraulic oil flow rate, l·min.$^{-1}$, and P is the hydraulic oil pressure, bar.

The fruit removal efficiency (η_{fr}) was calculated from Eq. (35):

$$\eta_{fr} = \frac{m_1}{m_2} \times 100 \tag{35}$$

Where η_{fr} is the olive fruit removal efficiency, %; m_1 the mass of harvested olive fruits from one tree, kg; and m_2 is the total olive fruits mass on the tree, kg.

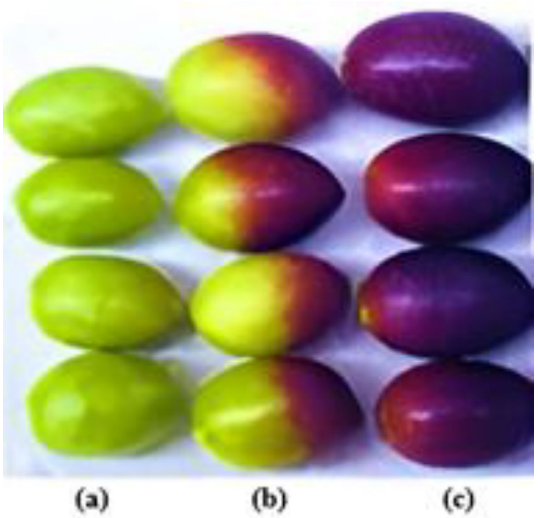


Fig. 4. Picual olive fruits at three ripening stages. (a) Full-mature, (b) Half-ripe, (c) Full-ripe.

Property	Full-mature		Half-ripe		Full-ripe	
	Mean value	CV[(b)]	Mean value	CV[(b)]	Mean value	CV[(b)]
Fruit length, mm	22.05 ± 0.14[a]	13.11	23.15 ± 0.08	12.18	24.81 ± 0.44	11.77
Fruit diameter, mm	13.71 ± 1.12	9.45	15.54 ± 0.17	8.15	17.09 ± 0.14	10.75
Stem length, mm	22.1 ± 0.34	1.67	22.3 ± 0.57	2.32	22.5 ± 0.37	2.15
Fruit mass, g	4.44 ± 0.12	10.34	5.22 ± 0.13	11.86	6.19 ± 0.12	11.46
Detachment force, N	7.4 ± 0.09	10.13	6.9 ± 0.13	9.46	6.6 ± 0.14	8.66
F/m, m s^{-2}	1666.7 ± 0.11	4.63	1321.8 ± 0.02	3.22	1066.2 ± 0.00	5.76
Trunk diameter, mm	78–186					

Table 2. Physical and mechanical characteristics of *Picual* olive. [a]Standard Error (SE); the difference between two means ≥ SE indicates a significant difference. [b]CV Coefficient of variation (Standard deviation divided by mean value).

Item	Full-mature	Half-ripe	Full-ripe	Average
Required power, kW	13.0	12.7	12.7	12.80

Table 3. The values of predicted required power (P_r).

Statistical analysis

The Standard Error (SE) was employed to detect significant deviations in treatment means. A multiple linear regression model was utilized to determine the relative contribution of related components to the dependent variable.

Plant guideline statement.

Experimental research and field studies on cultivated or wild plants, including the collection of plant material, comply with relevant institutional, national, and international guidelines and legislation.

Results and discussion

Plant characteristics of *Picual* olives

Table 2 presents the average measurements of olive fruit length, diameter, stem length, mass, detachment force, the ratio of detachment force to fruit mass (*F/m*), and trunk diameter. Properties such as fruit length, diameter, mass, and detachment force exhibited higher coefficients of variation (CV) exceeding 8%. Conversely, stem length and the *F/m* ratio displayed lower CV values below 6%. At the three ripening stages, all properties of olive fruit were statistically significant except for stem length.

Determination of the damping ratio (ζ)

Based on the information provided in Table 2 and Eq. (33), the damping ratios (ζ) were calculated. The average damping ratio (ζ) across all ripening stages was found to be 0.106.

Determination of the shaking frequency (ω)

Regarding the information in Tables 1 and 2, & 3 and Eq. (31), the values of shaking frequency (ω) are estimated. From Eq. 31, it is noted that the value of (ω) exists on both sides of the equation. Therefore, to determine the value of (ω), the trial and error method was used by substituting an imposed value for the (ω) on the right side of Eq. (30) and then calculating the value of (ω) from the left side. For the two values of (ω) to match, a correction factor (cf.) must be used. The values of cf. were 0.75135, 0.77463, and 0.77681 for full-mature, half-ripe, and full-ripe olive, respectively. Thus, the value of shaking frequency (ω) at any ripening stage was 18 Hz (113.1 rad·s^{-1}).

Theoretical deterioration of the predicted required power (*Pr*)

Using the data from Tables 1, 2 and 3, and 4, along with Eq. (32), the predicted required power (*Pr*) values were estimated and tabulated in Table 3. The average *Pr* values were calculated to be 13.0, 12.7, and 12.7 kW for the full-mature, half-ripe, and full-ripe olives, respectively. Overall, the average *Pr* value across all stages was 12.80 kW.

The actual shaking power (*Pa*)

The values of the actual shaking power (*Pa*) practically measured in the field are shown in Fig. 5. The shaking power values increased when the trunk diameter increased, and the power decreased as the clamping height increased. A larger trunk diameter means a stiffer and stronger tree trunk that needs more power to vibrate. The power also decreases as the clamp attachment height increases because as the shaker clamp is hinged higher up the trunk the distance between the clamp and the point of fixation of the tree (the point of the tree trunk when it contacts the soil) is greater thus to have the same oscillation as lower hinged clamps the required force is less, and the needed power is consequently less.

Olive trunk diameter, mm	Attachment height, mm					Average
	400	500	600	700	800	
78	89.7 ± 2.4[a]	90.1 ± 2.4	89.9 ± 0.9	90.6 ± 2.0	91.0 ± 1.6	90.3
84	89.6 ± 0.6	90.0 ± 1.6	89.5 ± 0.8	90.0 ± 0.9	90.2 ± 1.4	89.9
101	89.2 ± 0.9	89.8 ± 0.9	89.6 ± 1.4	90.0 ± 1.0	89.8 ± 0.7	89.7
143	88.3 ± 1.2	88.7 ± 1.3	88.0 ± 1.6	88.9 ± 2.2	89.4 ± 0.9	88.6
158	88.3 ± 1.4	88.3 ± 2.8	88.7 ± 0.8	88.9 ± 1.5	88.9 ± 1.2	88.6
186	88.2 ± 1.3	88.3 ± 1.7	88.9 ± 1.3	88.9 ± 1.6	88.9 ± 1.6	88.6
Average	88.9	89.2	89.1	89.5	89.7	89.3

Table 4. Fruit removal efficiency at different attachment heights and olive trunk diameters. [a]Standard Error (SE); the difference between two means ≥ SE indicates a significant difference.

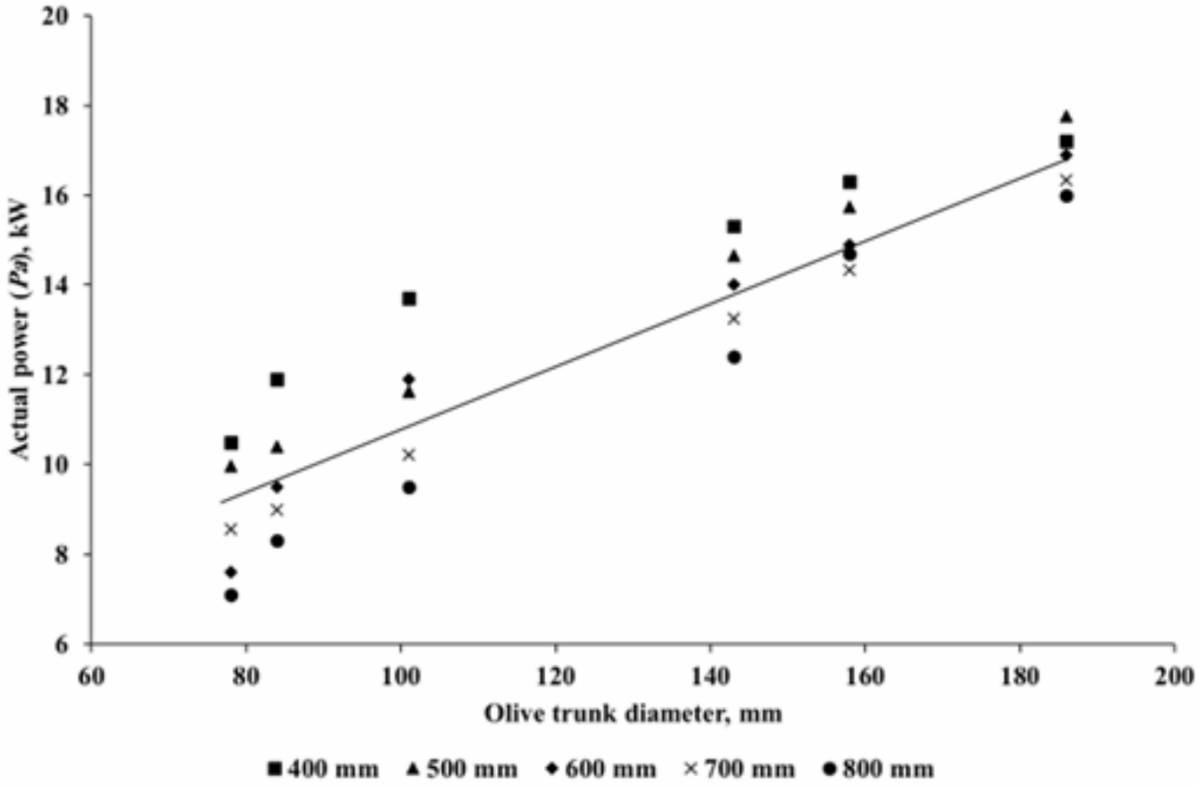


Fig. 5. The actual shaking power at different attachment heights and trunk diameters.

For 400 mm attachment height, the values of *Pa* were 10.5, 11.9, 13.7, 15.3, 16.3, and 17.2 kW at 78, 84, 101, 143, 158, and 186 mm trunk diameter respectively. For 500 mm attachment height, the values of *Pa* were 10.0, 10.4, 11.6, 14.7, 15.8, and 17.8 kW at 78, 84, 101, 143, 158, and 186 mm trunk diameter, respectively. For 600 mm attachment height, the values of *Pa* were 7.6, 9.5, 11.9, 14.0, 14.9, and 16.9 kW at 78, 84, 101, 143, 158, and 186 mm trunk diameter, respectively. For 700 mm attachment height, the values of *Pa* were 8.6, 9.0, 10.2, 13.3, 14.3, and 16.3 kW at 78, 84, 101, 143, 158, and 186 mm trunk diameter, respectively. For 800 mm attachment height, the values of *Pa* were 7.1, 8.3, 9.5, 12.4, 14.7, and 16.0 kW at 78, 84, 101, 143, 158, and 186 mm trunk diameter respectively. Multiple regression analysis was done between independent variables (attachment height "h" and trunk diameter "D_t") and the dependent variable, actual shaking power (*Pa*). The multiple R and R^2 were 0.983 and 0.966, respectively. The suitable model for predicting the *Pa* for the trunk shaker was:

$$Pa = -0.007042659h + 0.072132264D_t + 7.860065821 \quad (36)$$

The derived mathematical Eq. (32) does not include the attachment height (h) and the trunk diameter (D_t) of the olive tree. Therefore, the multiple regression method by the least squares method was used to find a statistical relation between the predicted required power (P_r) values calculated from Eq. (31) and actual values of the

required power as a function of the clamp attachment height (h) and the trunk diameter (D_t). The resulting equation was in the form:

$$Pa = P_r \times (0.618861142 - 0.000554379\, h + 0.005689014 D_t) \quad (37)$$

Where Pa is the actual required power, kW; P_r is the predicted required power (calculated from Eq. 32), kW; h is the clamp attachment height, mm, and D_t is the trunk diameter, mm. The regression coefficient of the predicted equation was 93.6% for h values between 400 and 800 mm and D_t values between 78 and 186 mm, and the mean absolute error (MAE) of the prediction model was 0.5009 kW.

Fruit removal efficiency (η_{fr})

The values of η_{fr} at different attachment heights and trunk diameters are illustrated in Table 4. The η_{fr} decreased with increasing the attachment height and the trunk diameter. At 400 mm of attachment height, the η_{fr} decreased by 0.1, 0.6, 1.6, 1.6, and 1.7%, while the trunk diameter increased from 78 mm to 84, 101, 143, 158, and 186 mm, respectively. The same trend was observed at 500, 600, 700, and 800 mm attachment heights. The results indicated that the fruit removal efficiency (η_{fr}) ranged from 88.2 to 91.0%. The highest η_{fr} values were achieved when trunk shaker operated under optimal harvesting conditions. All η_{fr} treatments were found to be insignificant.

While these trends were not statistically significant, the η_{fr} remained consistently high across all treatments, indicating stable performance of the mechanical system. This consistency suggests that the harvesting system can maintain efficiency under various conditions. Although the differences are not statistically meaningful, they provide valuable practical insights that could inform future optimizations of harvesting systems. In other words, these results can be used as a guide to identify which conditions may need improvement to achieve higher efficiency in the future.

Discussion: Trunk shakers execute controlled vibrations, which must be adjusted based on the specific tree species and its architectural characteristics to detach fruits effectively[23]. Required power is an essential factor in operating trunk shakers due to the global trend of energy conservation. In this research work, a mathematical analysis was done to predict the required power of the trunk shaker. The derived equation correlates the required power (P_r) to shaking displacement, fruit detachment force to fruit mass ratio, stem length, damping ratio, unbalanced mass of tree shaker, and exciting shaking frequency. The results showed that the average damping ratio and shaking frequency values were 0.106 and 18 Hz at any ripening stage of *Picual* olive fruits. This result is similar to those found by Ghonimy et al.[24] and O'brien et al.[25]. According to Ghonimy et al.[24], the damping ratio values for full-mature, half-ripe, and full-ripe were 0.103, 0.103, and 0.106 respectively. O'brien et al.[25], found that the optimum frequency for harvesting olive fruits using a trunk shaker ranged between 12 and 24 Hz. The results indicated that the average value of predicted P_r was 12.8 kW while the measured power value was 12.6 kW. These results are similar to those found by Horvath and Sitkei[9], who reported that the required power ranged from 6 to 14 kW. The trunk shaker was tested at five levels of attachment height: 400, 500, 600, 700, and 800 mm, and six levels of olive trunk diameter: 78, 84, 101, 143, 158, and 186 mm. Results indicated that the actual required power ranged between 7.1 and 17.8 kW with an average of 12.7 kW due to differences in tree trunk diameters and different attachment heights. Also, the results indicated that the fruit removal efficiency (η_{fr}) for all treatments ranged between 88.2 and 91.0%. These results are similar to those found by Blanco-Roldán et al.[22], who reported that the η_{fr} ranged from 87.8 to 90.8%. Multiple regression using the least squares method was used to find a statistical relation between the theoretical values calculated from the theoretical equation and actual values of the required power as a function of the clamp attachment height and the trunk diameter. The regression coefficient of the predicted equation was 93.6% for attachment height values between 400 and 800 mm and tree trunk diameters values between 78 and 186 mm, and the mean absolute error of the prediction model was 0.5009 kW.

Conclusion

This paper concludes that all characteristics of the investigated *Picual* olive fruits were statistically significant with the different ripening stages, except for fruit diameter and stem length. The average damping ratio values at each ripening stage were also observed to be 0.106. The value of the shaking frequency at any ripening stage was 18 Hz. A new mathematical analysis was done to predict the required power of the olive trunk shaker. The mathematical analysis showed that the average required power for the olive trunk shaker is 12.80 kW, and it is affected by shaking displacement fruit detachment force, fruit mass, stem length, damping ratio, unbalanced mass of the tree shaker, and exciting shaking frequency. The actual required shaking power for using an olive trunk shaker ranged between 7.1 and 17.8 kW at five attachment heights and six levels of olive trunk diameter. Multiple regression was used to find a statistical relation between the theoretical values and actual values of the required shaking power as a function of the clamp attachment height and the trunk diameter.

Data availability

The datasets generated and/or analyzed of the current study are available from the corresponding author on reasonable request.

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Additional information
Correspondence and requests for materials should be addressed to M.G. or M.M.I.

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